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INTRODUCTION

The paper contains 68 pages, 1 book, 11 figures, 14 tables, 9 references, and 2 appendices.

This study aims to explore a subset of a yellow taxi trips dataset published by the New York City Taxi & Limousine Commission, extract various valuable pieces of information from it, and interpret its hidden factors, especially those related to trip prices. To achieve this, we employ five basic data analysis techniques (linear regression, principal component analysis, cluster analysis, contingency tables, and bootstrap) and delve into their details to accurately analyze the results and gain insights into the dataset structure.

All computations have been performed using Python 3.10.11 [1] in a Jupyter [2] environment. The printed version of the notebook, including the source code and raw outputs of all cells, is presented in Appendix B. To simplify the implementations, we used several typical data analysis and machine learning Python libraries: numpy [3], pandas [4], matplotlib [5], seaborn [6], scikit-learn [7], and scipy [8].

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1. DATASET

The dataset used in the study is a subset of a huge database of yellow taxi trips in New York City for August 2025, published by the New York City (NYC) Taxi & Limousine Commission (TLC) [9]. The TLC is an agency of the New York City government created in 1971 that licenses and regulates the medallion (yellow) taxis, for-hire vehicles, commuter vans, and paratransit vehicles industries, including app-based companies such as Uber and Lyft. Over 200,000 TLC licensees complete approximately 1,000,000 trips each day. To operate for hire, drivers must first undergo a background check, have a safe driving record, and complete 24 hours of driver training. TLC-licensed vehicles are inspected for safety and emissions at a TLC Inspection Facility. For policy researchers to observe changing trends in the industry and inform decisions made by the City, the organization publishes monthly trip data generated by the licensees. Typically, the data is available with a two-month delay to allow time for full vendor submissions to be received.

In this study, we focus on a database of yellow taxi trip data for August 2025, which contains 20 features for over 3 million trips. For the task, a random subset of 1,000 records and the 16 most meaningful features with no missing values was selected. The list of features and their descriptions is presented in Table 1.

Table 1. Features available in the dataset

Feature name	Description	Feature type
VendorID	A code indicating the TPEP provider that provided the record.	Categorical: 1 = Creative Mobile Technologies, LLC 2 = Curb Mobility, LLC 7 = Helix
passenger_count	The number of passengers in the vehicle	Quantitative
trip_distance	The elapsed trip distance in miles reported by the taximeter	Quantitative
PULocationID	TLC Taxi Zone, in which the taximeter was engaged	Categorical, 1 to 265
DOLocationID	TLC Taxi Zone, in which the taximeter was disengaged	Categorical, 1 to 265

payment_type	A numeric code signifying how the passenger paid for the trip	Categorical: 1 = Credit card 2 = Cash 3 = No charge 4 = Dispute
fare_amount	The time-and-distance fare calculated by the meter	Quantitative
extra	Miscellaneous extras and surcharges	Quantitative
mta_tax	Tax that is automatically triggered based on the metered rate in use	Quantitative but with only two values: 0 and 0.5
tip_amount	Tip amount – automatically populated for credit card tips; cash tips are not included	Quantitative
tolls_amount	Total amount of all tolls paid for the trip	Quantitative
improvement_surcharge	Improvement surcharge assessed on trips at the flag drop. The improvement surcharge began being levied in 2015	Quantitative but with only two values: 0 and 1
total_amount	The total amount charged to passengers. Does not include cash tips	Quantitative
congestion_surcharge	Total amount collected in a trip for the NYS congestion surcharge	Quantitative but with only two values: 0 and 2.5
Airport_fee	For pick up only at LaGuardia and John F. Kennedy Airports	Quantitative but with only four values: 0, 1.75, 5, and 6.75
cbd_congestion_fee	Per-trip charge for MTA's Congestion Relief Zone starting Jan. 5, 2025	Quantitative but with only two values: 0 and 0.75

The distributions of all variables available in the dataset are shown in Figure 1. All things considered, we have four totally categorical features (VendorID, PULocationID, DOLocationID, and payment_type) and five truly quantitative ones (trip_distance, fare_amount, tip_amount, tolls_amount, and total_amount) that will be of the most interest in the study.

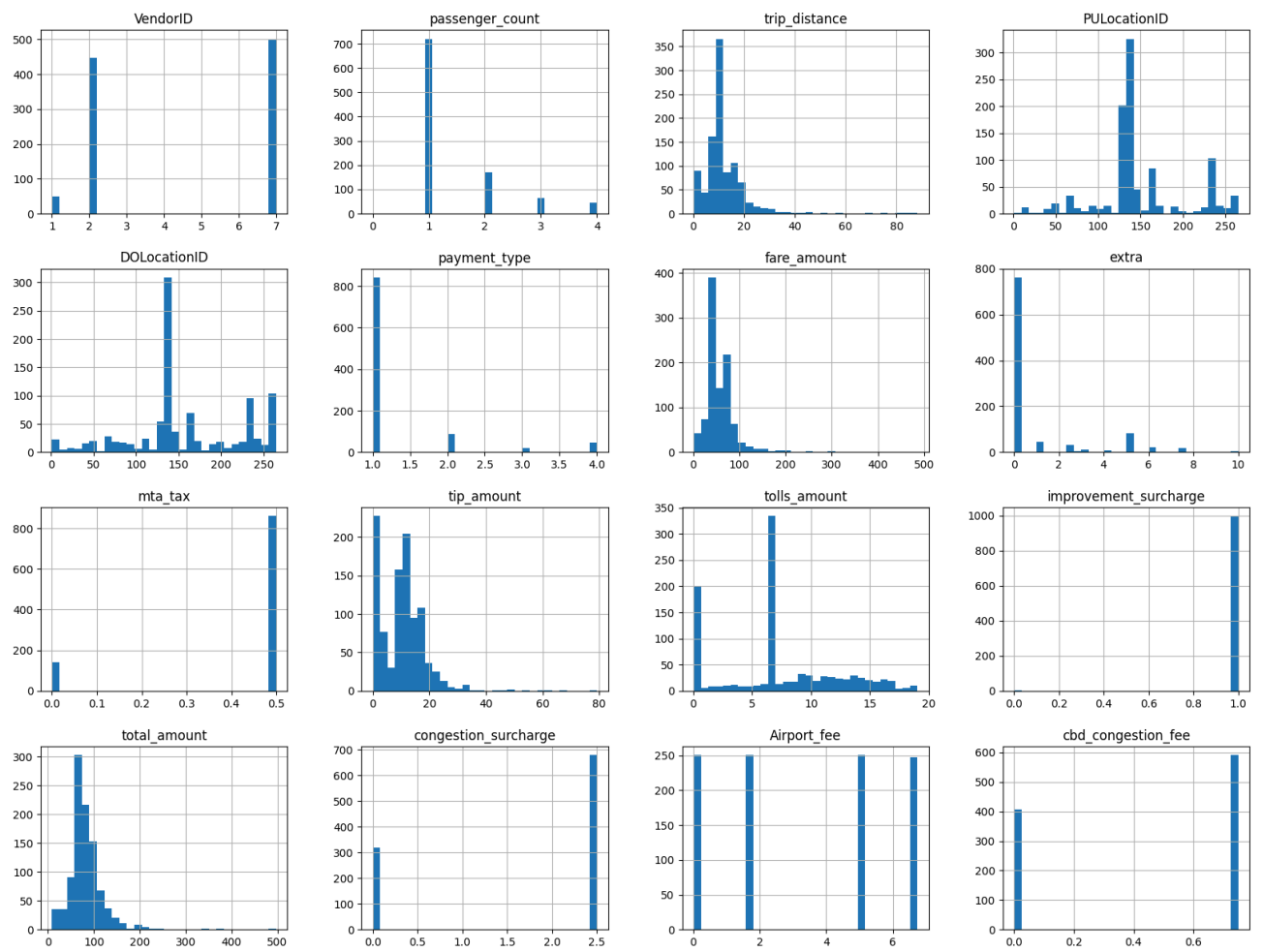


Figure 1. Distributions of all features available in the dataset

2. CORRELATION COEFFICIENT

For the first task, we need to explore the concept of correlation using two linear-like features from the dataset: analyze the scatter plot, build a linear regression, and comment on various metrics of its quality. For this, we will focus on the five truly quantitative features as outlined previously: trip_distance, fare_amount, tip_amount, tolls_amount, and total_amount.

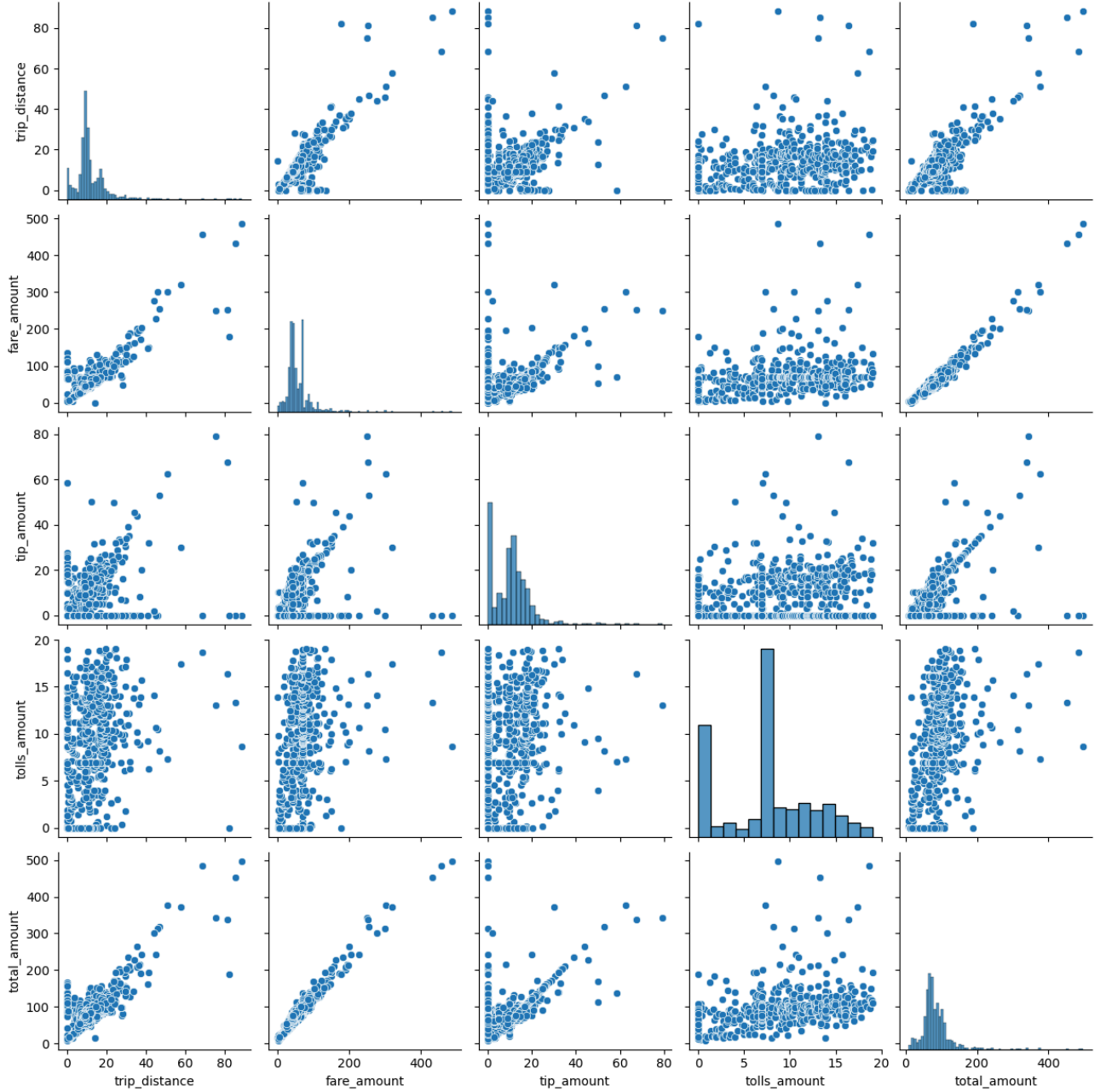


Figure 2. Pairplot of quantitative features

2.1. Selecting two “linear-like” features

From the pairplot shown in Figure 2, the two features with the most “linear-like” scatterplot are, expectedly, the `fare_amount` and `total_amount`. Further work will be carried out with these two variables.

2.2. Scatterplot of the selected features

The scatterplot of these features, shown in Figure 3, exhibits an outstandingly good linear dependency with no obvious outliers. Moreover, passing through or close to the origin with similar scales for both axes suggests that it may actually be a $y = x$ dependency. Therefore, the data are well-suited for building a linear regression.

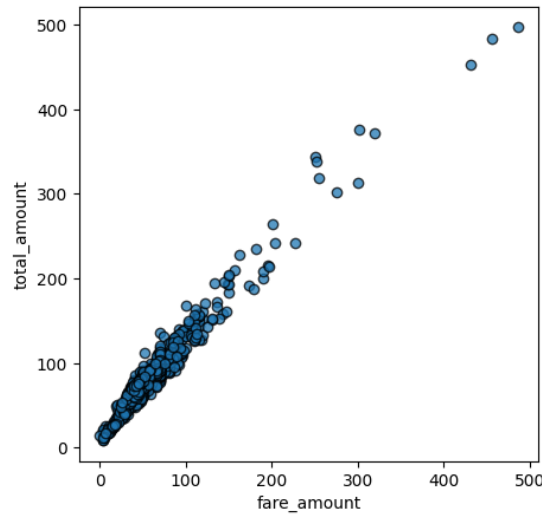


Figure 3. Scatterplot of features `fare_amount` and `total_amount`

2.3. Linear regression

Building a linear regression of `total_amount` over `fare_amount` results in equation (1) depicted over the original data in Figure 4.

$$\text{total_amount} = 1.083 \cdot \text{fare_amount} + 20.992 \quad (1)$$

As expected, the slope turned out close to one, and the intercept, although not zero, is small relative to the absolute values of the features. With a slope of 1.083, we know that on average, a

dollar added to “fare” results in one dollar and 8.3 cents added to the “total” price of the trip. Therefore, we may assume that “fare” is the primary regular factor contributing to the resulting cost, with other distance-based charges constituting only 8% of the price. The intercept of 20.992, on the other hand, indicates the amount of constant, fixed service fees charged to the passengers regardless of the trip fare.

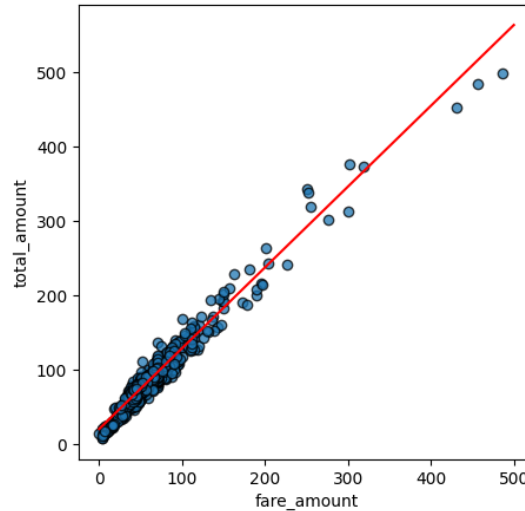


Figure 4. Scatterplot of features fare_amount and total_amount with a linear regression in red

2.4. Correlation and determinacy coefficients

The correlation coefficient between these features is 0.976, which is an outstanding result showing a remarkable linear relationship between the variables.

The determinacy coefficient is 0.953, which is also a great result confirming a robust linear relationship between fare and total amount. Formally, the value indicates that the linear regression explains around 95% of the variance of the total amount.

2.5. Predicting the target variable for four objects

To predict the target variable, we substitute the predictor’s value into equation (1) for four different trips. The obtained results are presented in Table 2. The predictions appear very accurate, with a relative error as low as 2% for object 2. This reinforces the previous assumptions that linear regression is a good fit for these features.

Table 2. Predictions of the target variable for four random objects

	Object 1	Object 2	Object 3	Object 4
Fare amount	95.0	136.7	70.2	52.7
True total amount	127.77	172.5	87.7	80.25
Predicted total amount	123.91	169.09	97.05	78.09
Absolute error	3.86	3.41	9.35	2.16
Relative absolute error (DA)	3.02	1.98	10.66	2.70
Relative absolute error (ML)	3.11	2.02	9.63	2.77

2.6. Predicting the target variable at all points of the dataset and comparing the errors to the determinacy coefficient

To further explore the results, we calculate the predictions for all trips in the dataset and compute the mean relative absolute error (MRAE), which turned out to be 11.32% according to data analysis, and 9.85% according to the machine learning view. Although far from zero, the errors are, on average, relatively low, indicating good quality of the predictions – the result consistent with what the determinacy coefficient suggests. Although high determinacy does not guarantee good prediction quality, the results indicate that it is the case for these two features.

Interestingly, the MRAE, according to machine learning (relative to the predicted value), is noticeably lower than the metric according to data analysis (relative to the true, observed value). This indicates that the linear regression tends to underpredict the target variable: that is, the prediction is typically smaller than the observed value.

Overall, linear regression is a good fit for the relationship between fare amount and the total price of yellow taxi rides in New York City, confirmed by multiple results we have obtained: high correlation and determinacy coefficients, small relative errors for both individual objects and the entire dataset.

3. PCA / SVD

For the second task, we need to compute the principal components of a subset of features to explore, visualize, and interpret the structure of the multidimensional data by reducing it to a lower dimension. Similar to the previous task, we will focus on the five quantitative features: trip_distance, fare_amount, tip_amount, tolls_amount, and total_amount. These variables describe the final cost of the trip and are highly correlated, which is the common aspect we will explore.

3.1. Computing principal components

To begin with, we apply singular value decomposition (SVD) to compute principal components for the dataset processed in four typical ways: no standardization, Z-scoring, rank standardization, and range standardization. The data scatter and the contributions of the principal components are presented in Table 3. The corresponding singular vectors (loadings) for all cases are presented in Tables 4 – 7.

As expected, the first principal component always contributes the most to the data scatter, while the last one is virtually useless, confirming the high correlation between the analyzed features. It's worth noting that for the original data, the first component explains almost 100% of the data scatter, while for the z-scoring, this metric accounts for only 63%, suggesting that the features have different scales (which is indeed true) that affect the results. Therefore, it is necessary to standardize the data, and we suggest Z-scoring as the most suitable way in our case.

Table 3. Data scatter and the contributions of all the principal components (naturally and percent) with four types of standardization

Standartization	Data scatter	PC-1	PC-2	PC-3	PC-4	PC-5
Original data	14'650'128 100%	14'452'991 98.7%	136'396 0.9%	31'830 0.2%	23'154 0.2%	5'757 0%
Z-Score	5'000 100%	3'151.1 63%	859.95 17.2%	780.58 15.6%	205.87 4.1%	2.5 0.1%
Range	106.57 100%	75.07 70.4%	20.53 19.3%	9.13 8.6%	1.83 1.7%	0.02 0%
Rank	330.61 100%	294.82 89.2%	24.63 7.4%	9.31 2.8%	1.83 0.6%	0.02 0%

Table 4. Singular vectors of all principal components for the original data

Standartization	PC-1	PC-2	PC-3	PC-4	PC-5
trip_distance	− 0.116	− 0.081	0.023	0.986	0.089
fare_amount	− 0.587	− 0.648	− 0.241	− 0.152	0.393
tip_amount	− 0.090	0.614	− 0.588	0.007	0.519
tolls_amount	− 0.062	0.176	0.750	− 0.067	0.631
total_amount	− 0.793	0.408	0.183	− 0.027	− 0.412

Table 5. Singular vectors of all principal components for the data after Z-scoring

Standartization	PC-1	PC-2	PC-3	PC-4	PC-5
trip_distance	0.494	0.083	0.350	− 0.791	− 0.014
fare_amount	0.533	0.081	0.264	0.469	− 0.648
tip_amount	0.283	− 0.816	− 0.469	− 0.114	− 0.144
tolls_amount	0.292	0.565	− 0.763	− 0.094	− 0.068
total_amount	0.554	− 0.034	0.076	0.363	0.745

Table 6. Singular vectors of all principal components for the data after range standardization

Standartization	PC-1	PC-2	PC-3	PC-4	PC-5
trip_distance	0.158	0.525	− 0.372	− 0.749	− 0.012
fare_amount	0.152	0.445	− 0.264	0.486	− 0.688
tip_amount	0.091	− 0.434	0.882	− 0.113	− 0.117
tolls_amount	0.952	− 0.300	0.043	− 0.031	− 0.022
total_amount	0.193	0.498	− 0.114	0.435	0.716

Table 7. Singular vectors of all principal components for the data after rank standardization

Standartization	PC-1	PC-2	PC-3	PC-4	PC-5
trip_distance	− 0.264	− 0.448	− 0.412	− 0.748	− 0.014
fare_amount	− 0.238	− 0.374	− 0.302	0.487	− 0.689
tip_amount	− 0.242	− 0.449	0.845	− 0.109	− 0.121
tolls_amount	− 0.85	0.524	0.028	− 0.029	− 0.025
total_amount	− 0.305	− 0.428	− 0.156	0.436	0.714

3.2. Interpreting principal components

To interpret the principal components, we consider the corresponding singular vectors presented in Tables 4 – 7 and restrict ourselves to the first two components, which explain the most variance in the data (more than 70% for all standardization methods). By the definition of PCA, each principal component is a linear combination of feature vectors with coefficients determined by the coordinates of singular vectors. Therefore, large absolute values of singular vector coordinates indicate big contributions of corresponding features to the respective principal component.

Considering the results for the original data, the first principal component is overwhelmed by the features “total_amount” and “fare_amount”, which is expected as these variables have a scale of around 10 times more than the other features, and, therefore, naturally explain the most variance in the dataset, which makes them more “important” during PCA. The second singular vector has the biggest positive values for the features “tip_amount” and “total_amount”, with a big negative value for the feature “fare_amount”. Therefore, the second principal component contrasts the fare amount with the tip amount and the total amount. However, this corresponds to less than 1% of the data scatter, confirming the obvious need to standardize the data to get meaningful interpretations.

Arguably, with Z-scoring, we observe the most interesting results, as mere scale is completely removed from the decisions. In this case, the first singular vector has significant values in all its coordinates, indicating that all five features contribute noticeably to the first principal component, with only the “tip_amount” and “tolls_amount” having a slightly smaller weight. This reinforces the previous observations that the five features are highly correlated and contribute to the data scatter together. Therefore, the first principal component reflects the general, primary part of the trip price. For the second singular vector, however, we obtain only two substantial coefficients: positive “tolls_amount” and negative “tip_amount”. Hence, the second principal component is designed to balance these two variables and capture an evident idea that higher toll costs during trips tend to reduce the amount of tips the passenger is willing to pay.

With range standardization, the scale is somewhat reintroduced as a factor in the algorithm, resulting in the first principal component being overwhelmed by the tolls_amount, which is an interesting result. Apparently, in this case, the amount paid for tolls is a significant factor that explains 70% of the data scatter virtually uniformly. For the second singular vector, all coefficients

are relatively big, meaning that the second principal component aims to contrast the obvious, regular fares (trip_distance, fare_amount, and total_amount) with “extra”, less typical charges (tip_amount and tolls_amount). Assuming the total amount consists of all other charges, this component essentially calculates the distance-based part of the fee, which corresponds to 20% of the data scatter.

With rank standardization, the PCA focuses even more on the tolls fee, which is the primary factor in the first component, explaining almost 90% of the data scatter, and the only positive coefficient in the second singular vector, contributing 7% to the data scatter.

Overall, the results strongly support the need for data standardization before applying PCA, as the original data did not yield positive results. Of the three typical standardization techniques, Z-scoring is the most robust, producing expected and interpretable outcomes, focusing on the total cost of the trip and the contrast between additional charges such as tolls and tips. The range and rank standardizations clearly suggest that tolls are a significant factor explaining the data scatter. Although this is an interesting result that may shed light on some patterns in the data, it is evidently not the most important observation to emerge from this analysis.

3.3. Visualizing the data using PCA

For visualization, we focus only on the first two principal components obtained after range standardization and Z-scoring, which, as expected from Table 3, explain 89.7% and 80.2% of the data scatter, respectively. For interpretability in the visualization, we explicitly categorize the dataset into multiple groups depicted in different colors using percentile groups 0 – 25, 25 – 75, and 75 – 100 for the three criteria we have seen previously: total amount, tip amount, and tolls amount, as well as an external factor – the vendor that reported the trip to the TLC.

From what we have seen previously, the first principal component after range standardization is overwhelmed by the tolls_amount feature, which is confirmed by the visualization in Figure 5: the third picture for the partition over percentile groups of the tolls amount shows clear separation of the three groups, with lower values of the first principal component corresponding to lower values of the tolls amount. The second principal component, in turn, was assumed to represent the distance-based part of the fee, which is the predominant element of the total amount. Accordingly, as shown in the first picture of Figure 5, the second principal component corresponds to the partition of objects over this feature: the lower the total amount, the lower the second principal component. For

the tip amount (picture 2 in Figure 5), there is no clear correlation between the partition and the principal components, which is consistent with our previous findings. However, there is an intriguing result for the partition by vendor. While CMT and CM are mostly mixed with no clear separation, the majority of trips reported by Helix are represented by two slightly slanted yellow lines. Accounting for the other findings, we may safely assume that most trips reported by Helix paid similar toll fees and may be clustered into two distinct groups. Apparently, most of Helix's operations are conducted along two routes that involve two different toll roads.

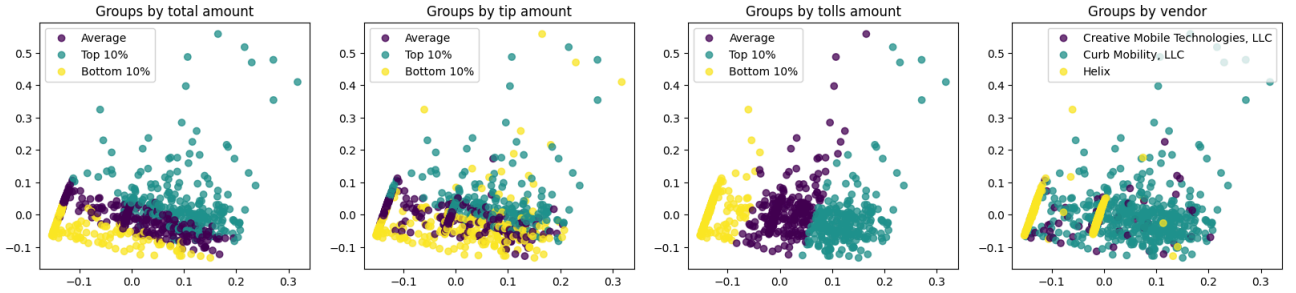


Figure 5. Scatter plot of the first two principal components of all objects after range standardization, with four categorizing factors depicted in different colors

For Z-scoring, we have previously concluded that the first principal component reflects the total price of the trip, which is precisely what we observe in the first picture of Figure 6: the percentile groups are positioned expectedly according to the first principal component, and, importantly, the lines separating the groups are almost horizontal. The second principal component, on the other hand, contrasts tips and tolls, which is specifically what we see in pictures 2 and 3 of Figure 6: both visualizations show three distinct groups of objects slanted in different directions, with the tip amount making the second principal component smaller, and the tolls amount generally making it bigger, which is exactly what one would expect from the singular vector having negative coefficient for `tip_amount` and positive coefficient for the `tolls_amount`. For the partition by the vendor shown in the last picture of Figure 6, there is no clear separation of groups in the visualization, with conceivably only Helix standing out from the rest. However, no interesting patterns similar to those observed in the results after range standardization are detected in this case.

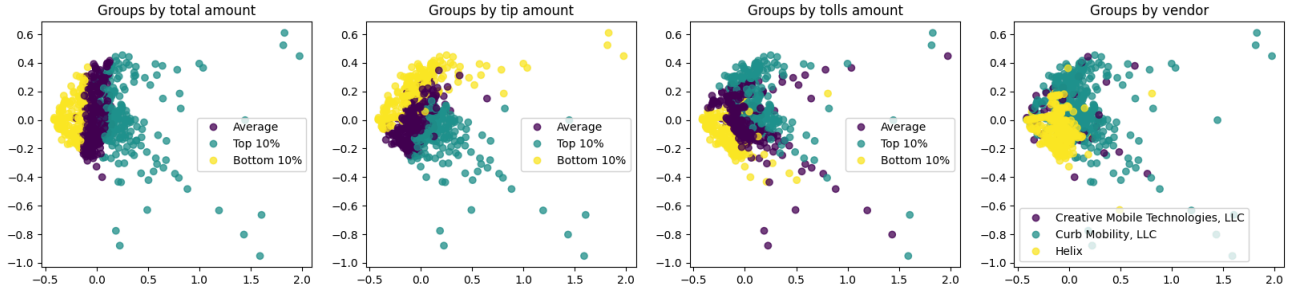


Figure 6. Scatter plot of the first two principal components of all objects after range standardization, with four categorizing factors depicted in different colors

3.4. Conventional PCA

Conventional PCA is another approach to computing principal components based on the covariance matrix instead of the singular value decomposition of the data. To explore this approach, we also compute the first two principal components and construct a similar visualization presented in Figure 7. As expected, the results are generally consistent with those presented in Figure 6 for the Z-scoring, although conventional PCA arguably produces less scattered components, with most of the data concentrated around a single point. The primary patterns from Z-scoring, though, persist in the conventional PCA: the first principal component reflects the total cost of the trip (see picture 1 in Figure 7), with the second component aimed to contrast tips and tolls (Figure 7, pictures 2 and 3), despite the separation of percentile groups over the toll amount being significantly less noticeable. For the partitioning by the vendor (Figure 7, picture 4), conventional PCA also produces no interesting results.

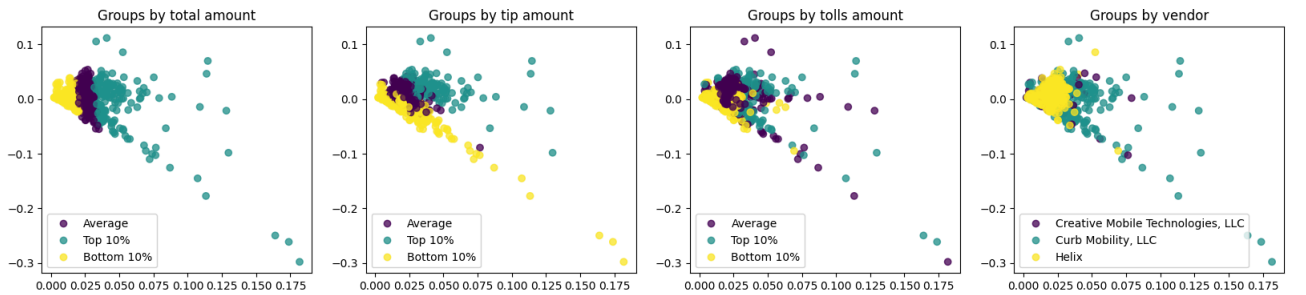


Figure 7. Scatter plot of the first two conventional principal components of all objects, with four categorizing factors depicted in different colors

Overall, Z-scoring provides the most interpretable results, which are generally supported by those of conventional PCA, making it the best approach to standardizing these data of the four

considered. Although other techniques may deliver some intriguing insights into the data, Z-scoring detects the most robust and consistent ones supported by a general understanding of the dataset. Moreover, as confirmed by the histograms in Figure 1, the five quantitative features considered resemble a normal distribution, which is precisely the case where Z-scoring is designed to show the best outcomes.

3.5. Hidden ranking factor

Principal component analysis also provides a framework for ranking the objects in the dataset according to some hidden factor, which, as we have observed previously when exploring the results for rank standardization, is the amount of tolls paid for the trip. After normalizing the first singular vector, the ranking criterion r should be computed according to equation (2). Clearly, the toll amount is the primary contributing factor, with the other four features following behind with almost equal weights. In other words, the most expensive trips are those with big charges in all categories, with special focus on toll fees.

$$r = 0.139 \cdot \text{trip_distance} + 0.125 \cdot \text{fare_amount} + 0.127 \cdot \text{tip_amount} + 0.448 \cdot \text{tolls_amount} + 0.161 \cdot \text{total_amount} \quad (2)$$

This result is confirmed by the explicit view of the most and least scored objects in the dataset displayed in Figure 8. Consistently, the highest-ranked trips are those with large tolls and moderate or high values of other charges. The objects with the lowest score, on the other hand, are short trips with no tolls, generally with no fees, and a very short distance.

trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	ranking	trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	ranking
68.61	455.9	0.00	18.66	484.06	81.999781	0.00	9.3	3.51	0.0	21.06	1.215890
81.38	252.0	67.53	16.36	337.64	79.406640	0.95	6.5	2.95	0.0	17.70	1.092769
57.87	319.4	30.00	17.38	372.03	74.934990	0.79	5.8	2.66	0.0	15.96	0.945807
75.19	250.0	79.21	13.04	343.25	72.641550	1.96	9.3	0.00	0.0	20.05	0.926309
85.19	431.5	0.00	13.30	452.55	70.345927	2.05	10.0	0.00	0.0	16.50	0.841936
24.79	133.9	32.14	19.06	194.60	63.423960	1.37	9.3	0.00	0.0	17.55	0.751457
88.35	486.0	0.00	8.69	497.44	62.884845	0.00	3.0	0.00	0.2	8.95	0.561522
29.92	150.0	33.78	17.88	204.41	62.457659	0.68	5.1	0.00	0.0	11.60	0.339453
33.93	162.6	45.37	14.88	228.60	59.020079	0.00	4.4	0.00	0.0	12.65	0.248975
37.82	203.9	20.00	15.68	242.33	58.945404	0.00	3.7	0.00	0.0	11.95	0.207966

Figure 8. Top-10 highest-ranked (left) and lowest-ranked (right) objects in the dataset, with the ranking factor (0 to 100) presented in the last column

Finally, we examine the relations between the ranking factor and the five features considered throughout this section. Figure 9 confirms that the toll amount is indeed the primary factor, with an almost perfectly linear dependency on the ranking factor. The remaining features primarily exhibit similar distributions, with only the tip amount slightly standing out due to the horizontal line at the bottom of the chart. Apparently, tipping is a less significant contributor to the overall cost of the trip, which is to be anticipated since tips are expected to be more common on shorter trips.

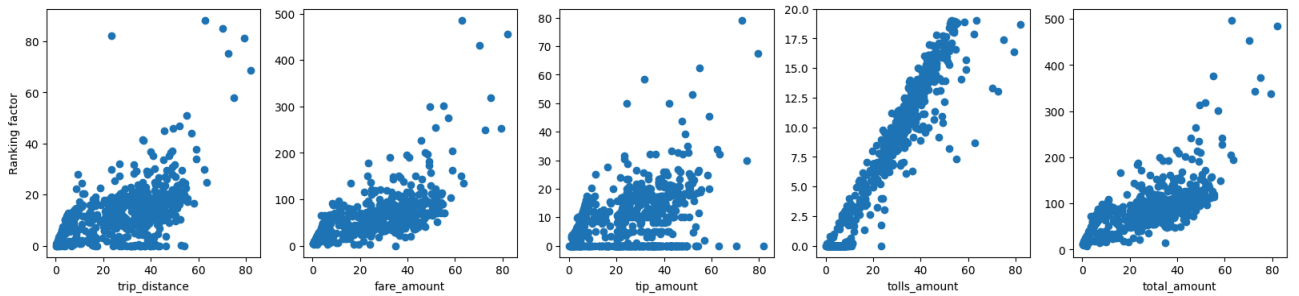


Figure 9. Scatterplot of the ranking factor with respect to the five quantitative features considered

4. CLUSTER ANALYSIS AND CLUSTER INTERPRETATION

For the cluster analysis, we drop the features PULocationID and DOLocationID and apply one-hot encoding to the other categorical features – VendorID and payment_type. To construct the clusters, however, we employ only the five quantitative features similar to the previous task: trip_distance, fare_amount, tip_amount, tolls_amount, and total_amount. These features are related to the total price of the trip, which is the common factor we expect to observe in the clusters. The remaining features will be used solely for the interpretation of the results, and will not participate in the construction of the clusters.

For the analysis, we will employ the K-Means algorithm, which operates by randomly initializing the requested number of cluster centers and proceeds to iteratively relabel the objects according to the closest cluster center and recalculate the new centers until convergence. Since the technique heavily relies on the distances between objects, it is necessary to standardize the data to make the features comparable. For this, we applied Z-scoring, which demonstrated the best results in the previous section. For completeness, we have also briefly validated the range standardization, which did not lead to significant improvements in the interpretability of the results and, thus, will not be considered further.

4.1. Applying K-Means and choosing the best random initialization

To start with, we apply K-Means to partition the objects into 4 and 7 clusters. For numerical stability, we perform 10 runs of the algorithm with different states of the random initialization and select the one with the smallest inertia criterion value for further analysis. The inertia criterion is computed as the sum of squared distances of samples to their closest cluster center at the last iteration of the algorithm and represents the quality of the partition, that is, how compact the obtained clusters are. Therefore, the smaller the inertia is, the more compact the clusters are, and the better the partition is considered to be.

The results for all runs of the algorithm with 10 preselected values of the random state initialization are shown in Table 8. Clearly, the best run for 4 clusters used a random state of 1, and the best partition into 7 clusters operated with a random state of 7, which are the values used further throughout this section for cluster analysis.

Table 8. Inertia criterion values for the 10 runs of K-Means with 4 and 7 clusters at random initializations. The best result for each number of clusters is highlighted in green

Random state	4 clusters	7 clusters
0	2085.95	1472.14
1	2074.59	1412.23
7	2085.94	1384.83
10	2085.95	1396.73
42	2085.94	1487.04
750	2085.94	1394.67
1234	2091.54	1439.09
2025	2085.94	1421.30
2026	2091.61	1423.74
3058	2189.40	1394.69

4.2. Interpretation of clusters

To gain insights into the obtained partitions, we calculate the relative differences between within-cluster means and the grand means (calculated over the full dataset) of each feature, including those not used for cluster construction but available in the original dataset. The raw results are presented in Appendix A. We expect to see clear distinctions between clusters in some or even most features, indicating common patterns of objects from one cluster that are not observed in objects from other clusters. Moreover, we anticipate these distinctions to be related to the hidden factor we assumed previously, which is the total price of the trip.

4.2.1. 4 clusters

Considering the 4-cluster partition presented in Table A1, we have one small cluster with long, costly trips, one giant cluster with an average price, and two moderately sized clusters with trips that cost slightly below and above average, respectively.

For the minority cluster 0, consisting of only 13 objects, we observe a massive positive relative difference for all quantitative features involved in constructing the partitions, indicating that these objects have significantly higher values of these characteristics than the average – these are the longest and most costly trips with huge tips from the passengers. Considering other features (not involved in K-Means), one may also notice several interesting patterns. Specifically, these trips had much lower MTA (Metropolitan Commuter Transportation Mobility) taxes than the average, no

congestion surcharge, and a small CBD congestion fee, suggesting that these trips were generally far from the city center and did not pass through the most expensive region of NYC – Manhattan, which is expected for long commutes. Interestingly, these trips were mostly operated by Corb Mobility LLC, and cash was the primary payment method used by the passengers.

The majority cluster 1, consisting of 540 objects, exhibits all relative differences within 20%, indicating that these are the most typical trips. Interestingly, this is the only cluster with a positive relative difference for a “Dispute” payment method – apparently, on average, people argue about the payment. Likewise, these trips were mostly reported by the two companies other than Creative Mobile Technologies LLC, meaning that the latter focuses on trips that slightly deviate from the average.

For cluster 2 (252 objects), all quantitative features exhibit around a -30% relative difference, indicating that these are slightly shorter trips than average. Expectedly, these trips also had noticeably smaller toll fees than average, as toll roads are generally not used for short trips. These trips were mostly reported by Helix and Creative Mobile Technologies LLC, with Helix focusing especially on such trips. From the payment methods, “no charge” is clearly prevalent, which could have been anticipated for short trips, and disputes rarely happen in these cases. Interestingly, this cluster is also the only one with a positive relative difference for the airport fees, indicating that most trips to NYC airports are shorter than average, which is consistent with the two primary airports (JFK and LaGuardia) being near the city center.

Finally, cluster 3 with 195 objects consists of trips that are slightly above average in terms of distance and fees, which is one of the few logically remaining options. These trips rarely involve airport fees and are primarily reported by Corb Mobility LLC, with Helix being the least likely to report a trip from this cluster. As expected, these trips have a less-than-average probability of having a “no charge” payment method, with the most common payment method being a credit card.

4.2.2. 7 clusters

In the case of a 7-cluster partition presented in Table A2, we generally observe the same trends: the clusters are ordered by the total price of the trip, with some clusters, however, suggesting more detailed patterns.

First, cluster 6, consisting of only 8 objects, corresponds to the minority cluster from the previous partitioning, with the same conclusions: these are the long, expensive trips reported primarily by Corb Mobility LLC, low MTA tax, no congestion surcharge, and cash as the primary payment method.

Going further, cluster 1 with 27 objects is a new group not previously recorded, described by slightly smaller yet still far above-average charges exhibiting similar patterns. Interestingly, these are usually reported by the Creative Mobile Technologies LLC, which is, in fact, consistent with the above-average cluster we have seen before.

Next, clusters 2 (162 objects) and 4 (169 objects) represent trips with distances and charges slightly above average, with the primary distinction being the tip amount: cluster 2 contains trips with below-average tips, while cluster 4 collects those with above-average tips. Considering the additional features, trips with low tips (cluster 2) have high extra charges, which is logically consistent: the passengers are unlikely to leave a tip if they have already been charged for miscellaneous extra services. Moreover, there is a clear difference in the payment methods between these groups: trips with tips are mostly paid by credit cards, while this is the single least popular option when passengers do not leave tips.

Subsequently, clusters 3 (353 objects) and 5 (175 objects) represent the objects with slightly below-average fares, with the major distinction being the toll fees, which are around-average for cluster 3 and noticeably below-average for cluster 5. Consistent with the results for a 4-cluster partition, these are the only clusters with above-average airport fees, indicating that most trips involving airports fall into one of these groups.

Finally, cluster 0 (106 objects) consists of the cheapest and shortest trips, which is the only logically sound option remaining. Creative Mobile Technologies LLC mostly reports these trips, and the prevalent payment methods are cash and “no charges,” which is also consistent with our previous findings.

4.2.3. Conclusion

Overall, both partitions are highly interpretable and, most importantly, provide consistent results based on the total price of the trips. Nonetheless, the larger partition (7 clusters) seems slightly overwhelming, with many repetitive interpretations that focus on specific patterns rather

than generalizing the data. Conclusively, we consider the 4-cluster partition more interpretable as it provides a clearer clustering outcome, better reflects the general trends, and does not delve into overly detailed concepts that are not interesting for the high-level overview of the data.

5. CONTINGENCY TABLE

Next, we construct and analyze contingency tables to explore the conceptual relations in the data – whether two nominal features can be considered statistically independent – which may uncover interesting associations in the data.

5.1. Creating nominal features

For this section, we focus on the dependencies between a categorical feature “passenger_count” and a quantitative feature “trip_distance”.

The former is inherently nominal and does not need additional preprocessing for further computations.

The latter, on the other hand, needs to be developed into the nominal form by binning the values using some predetermined thresholds. Based on the deep minima clearly observable in Figure 10, we suggest the following bins: $[0; 4)$, $[4, 14)$, $[14, 22)$, $[22, \infty)$ corresponding, respectively, to short trips, average trips, long trips, and very long trips, similar to what we found previously when exploring the clusterization.

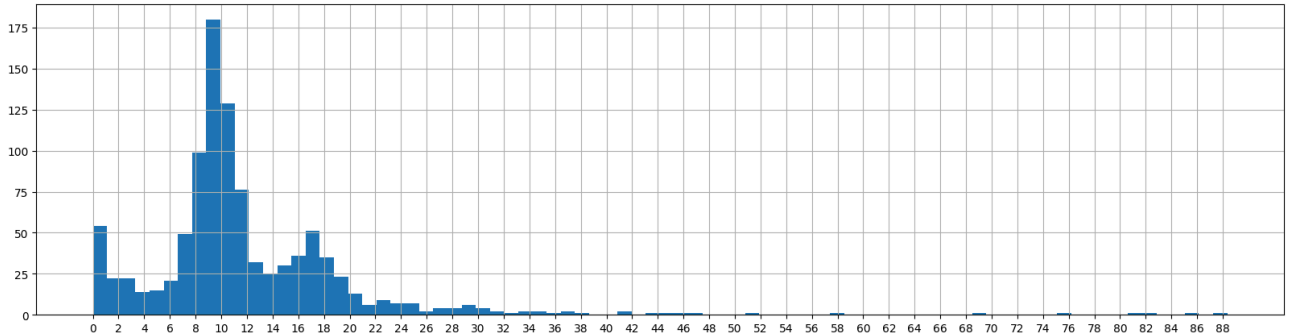


Figure 10. Histogram of the distribution of trip_distance

As a result, we have two nominal features with 5 unique values for passenger_count – 0, 1, 2, 3, 4 (interestingly, the dataset contains one trip without passengers), and 4 unique values for trip_distance.

5.2. Contingency table, conditional frequency, and Quetelet index tables

To construct the contingency table, one needs to partition the dataset based on the values of the two features and compute the size of each group. The respective result is shown in Table 9.

Then, we define relative frequency by dividing each cell of the contingency table by the number of objects in the dataset, and compute the conditional frequencies presented in Table 10 as the quotient of the relative frequency over the column-sum for the respective category of trip_distance. Finally, we compute the Quetelet index (Table 11), which shows the relative change of the probability of passenger_count under conditions of trip_distance.

Table 9. Contingency table of features passenger_count and trip_distance

	[0; 4)	[4; 14)	[14; 22)	[22; ∞)	
0 passengers	0	0	1	0	1
1 passenger	64	478	131	48	721
2 passengers	16	105	39	9	169
3 passengers	12	26	19	6	63
4 passengers	14	15	12	5	46
	106	624	202	68	1000

Table 10. Conditional frequency of feature passenger_count over trip_distance

	[0; 4)	[4; 14)	[14; 22)	[22; ∞)	
0 passengers	0.00	0.00	0.00	0.00	1
1 passenger	0.60	0.76	0.65	0.71	721
2 passengers	0.15	0.17	0.19	0.13	169
3 passengers	0.11	0.04	0.09	0.09	63
4 passengers	0.13	0.02	0.06	0.07	46
	106	624	202	68	1000

Table 11. Quetelet index of features passenger_count and trip_distance. The most meaningful values are highlighted in bold

	[0; 4)	[4; 14)	[14; 22)	[22; ∞)	
0 passengers	-1.00	-1.00	3.95	-1.00	1
1 passenger	-0.16	0.06	-0.10	-0.02	721
2 passengers	-0.11	-0.00	0.14	-0.22	169
3 passengers	0.80	-0.34	0.49	0.40	63
4 passengers	1.87	-0.48	0.29	0.60	46
	106	624	202	68	1000

Expectedly, most trips operate with one or two passengers, and medium-distance trips (4 to 14 miles) clearly predominate. The largest Quetelet index is observed for the long trips with 0

passengers, short trips with 3 or 4 passengers, and very long trips with 4 passengers, which implies that these trips occur noticeably more often than average. Nonetheless, the results are very noisy due to a small number of objects in those categories, and should be treated with great caution. Similarly, extreme negative values for almost all categories with 0 passengers are practically meaningless, as there is only one such trip in the dataset. For the largest categories, on the other hand, the Quetelet indices are relatively small, indicating that those pairs are around as frequent as the average. Yet, some values are still around 0.1, which is relatively significant to assume meaningful relations between the passenger count and the trip distance.

5.3. Average Quetelet index and Pearson's chi-squared

To verify this hypothesis, we compute the average Quetelet index and Pearson's chi-squared, which are both approximately equal to 0.0487, indicating that knowledge of passenger count adds around 5% to the frequency of trip distance. Although the value is quite low, it suggests some dependence between the features and requires more rigorous tests to assert their statistical independence.

It must be mentioned that the Quetelet index and Pearson's chi-squared, although computed differently, are precisely equal, as they both estimate the same aspect of data in slightly varying ways. Both metrics summarize how much the joint frequencies deviate from what would be expected under independence, evaluate the overall strength of association between the two nominal features, and are somewhat measures of the correlation between them.

5.4. Minimum number of observations for independence

Finally, we perform a chi-squared statistical independence test on our data at confidence levels of 95% and 99%. Considering that the analyzed features contain 4 and 5 unique values, the theoretical distribution of the density function is the chi-squared distribution with $(4 - 1) \cdot (5 - 1) = 12$ degrees of freedom. The critical values at 95% and 99% for this distribution are 21.02 and 26.22, while the observed value in our data is $1000 \cdot 0.0487 = 48.7$. Therefore, the hypothesis of statistical independence between these two features should be rejected at both the 95% and 99% confidence levels, which is consistent with the previous results. Moreover, $\frac{21.02}{0.0487} \approx 432$ and $\frac{26.22}{0.0487} \approx 538$ observations would be sufficient to see the features as associated at 95% and 99% confidence levels, respectively.

6. BOOTSTRAP

Finally, we revisit the results from clustering analysis and explore whether the means in two clusters (going further, we focus on the most numerous clusters 1 and 2 from the 4-cluster partition) coincide with each other and with the grand mean of the dataset using bootstrapping techniques. For this, we focus on the `total_amount` feature, which, as we have seen previously and confirm in Figure 11, effectively distinguishes the clusters. Therefore, we expect to verify that the means of the feature in different clusters are indeed statistically different.

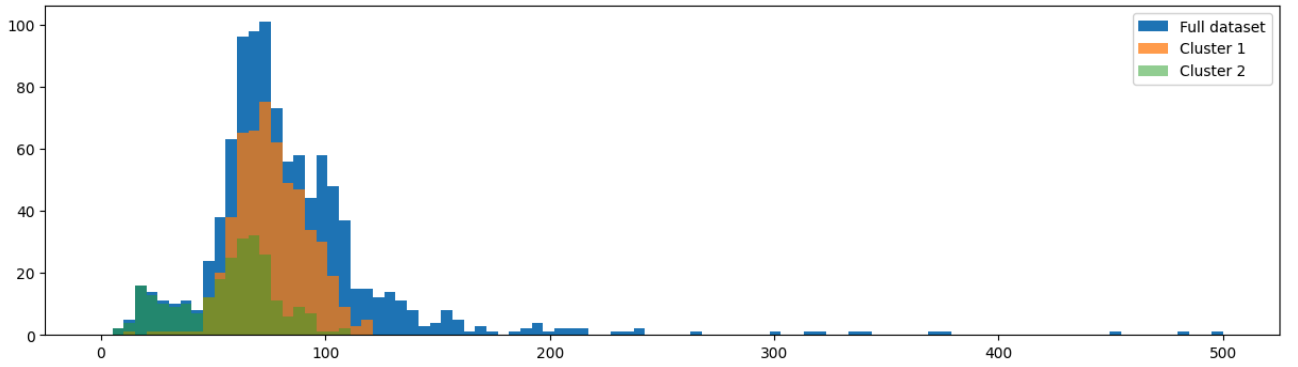


Figure 11. Histograms of the distribution of `total_amount` in the full dataset and in clusters 1 and 2 of the 4-cluster partition

For bootstrapping, we generate 10,000 random samples from the data with replacement and compute the mean within each sample. According to the central limit theorem, the distribution of those means is assumed to be normal, allowing us to estimate the confidence intervals of the metric by taking the symmetric quantiles of the normal distribution (pivotal method) or the percentiles of the sequence (non-pivotal method) as the boundaries. Following this section, all confidence intervals are presented at a 95% confidence level. For comparison, we also report confidence intervals, assuming that the feature is normally distributed where possible.

6.1. Estimating the grand mean

First, we calculate the confidence intervals for the grand mean as described previously. Additionally, Table 12 reports similar results for the within-cluster means of both clusters, which will be necessary in the following parts of this section.

The intervals obtained using the two versions of bootstrapping are similar yet totally differ from the results under the Gaussian assumption, which is the expected result, as Figure 11 clearly

indicates that the `total_amount` is not distributed normally for either the full dataset or any of the clusters, thus making the assumption incorrect.

Considering the grand mean, with 95% confidence, we may assume that it is between approximately 81.6 and 87.2, which is consistent with the sample mean of the data (84.39). Moreover, the sample mean is extremely close to the centers of the confidence intervals, which is 84.37 in both the pivotal and non-pivotal versions.

Table 12. 95% confidence intervals for the grand mean and within-cluster means of `total_amount`

	Grand mean	Cluster 1	Cluster 2
Under the Gaussian assumption	[− 3.28; 172.05]	[45.67; 106.87]	[13.26; 98.98]
Pivotal bootstrap	[81.61; 87.14]	[74.95; 77.59]	[53.42; 58.81]
Non-pivotal bootstrap	[81.69; 87.25]	[74.93; 77.57]	[53.41; 58.82]

6.2. Comparing the means in two clusters

To compare the means of the feature in two clusters, following the idea of bootstrapping, we compute the difference of the means between two random samples from each cluster, and calculate the 95% confidence intervals for that difference using both the pivotal and non-pivotal approach.

As shown in Table 13, the intervals are generally similar for both versions of bootstrapping. Most importantly, they undoubtedly do not include zero, indicating that the hypotheses of similarity of the means in the two clusters should be rejected at a 95% confidence level, which is consistent with our previous findings.

Table 13. Bootstrap mean with its 95% confidence intervals for the difference between the within-cluster means of `total_amount`

	Mean difference	95% confidence interval
Pivotal bootstrap	20.18	[17.19; 23.18]
Non-pivotal bootstrap		[17.18; 23.15]

6.3. Comparing a within-cluster mean with the grand mean

For the last part of this section, we focus on one of the clusters and evaluate whether its mean is similar to the grand mean of the whole dataset. For this, we focus on the first cluster, which, according to Table 12, has its within-cluster mean closer to the grand mean, supported by the

non-intersecting confidence intervals for the two clusters. Similar to the previous part of this section, we compute the differences between the means of a random sample from the full dataset and a random sample from the first cluster, and report the 95% confidence interval for that difference in Table 14.

Table 14. Bootstrap mean with its 95% confidence intervals for the difference between the grand mean and the cluster-1 mean of total_amount

	Mean difference	95% confidence interval
Pivotal bootstrap	8.10	[5.05; 11.15]
Non-pivotal bootstrap		[5.13; 11.26]

Again, the obtained confidence intervals for both versions of the bootstrap coincide and certainly do not include zero, indicating that the hypotheses of equality between the grand mean and the cluster-1 mean must be rejected at 95% confidence level.

CONCLUSION

This study analyzes a small subset of the yellow taxi trips published by the New York City Taxi & Limousine Commission using five basic techniques from data analysis. Despite the small amount of data and the relative simplicity of the methods used, we revealed several interesting patterns across all sections of the paper that provide important insights into how taxis operate in New York City. Many of the outcomes have been confirmed robust and were consistent with the common knowledge of the business and the layout of NYC. Undoubtedly, one can expand the study to the full dataset published by the TLC to identify even more interesting patterns, observe changing trends in the industry, and conceivably inform decisions made by the City, which is the primary purpose of this dataset.

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Appendix A

Table A1. Cluster interpretation for K-Means with 4 clusters. The most outstanding results, referenced in the main text, are highlighted in green

		trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	passenger_count	extra	mta_tax	improvement_surcharge	congestion_surcharge	Airport_fee	cbd_congestion_fee	Vendor			payment_type			
														Creative Mobile Technologies, LLC	Curb Mobility, LLC	Helix	Credit card	Cash	No charge	Dispute
Grand mean		11.93	58.52	10.41	7.38	84.39	1.43	0.97	0.43	0.99	1.7	3.36	0.44	0.05	0.45	0.50	0.84	0.09	0.02	0.05
Cluster 0: 13 obj.	Cluster mean	62.06	303	25.98	11.32	345	1.54	2.35	0.12	1.00	0.00	1.87	0.17	0.00	0.92	0.08	0.69	0.31	0.00	0.00
	Difference	50.13	244	15.58	3.94	261	0.11	1.38	-0.31	0.00	-1.70	-1.50	-0.27	-0.05	0.48	-0.43	-0.15	0.22	-0.02	-0.05
	Relative difference	420	417	150	53.42	309	7.43	143	-73.17	0.20	-100	-44.51	-61.02	-100	106	-84.7	-17.8	246	-100	-100
Cluster 1: 540 objects	Cluster mean	10.06	50.13	8.77	8.69	76.27	1.38	1.01	0.44	1.00	2.00	3.56	0.47	0.03	0.41	0.56	0.82	0.09	0.02	0.06
	Difference	-1.87	-8.38	-1.64	1.31	-8.11	-0.05	0.05	0.01	0.00	0.30	0.20	0.02	-0.02	-0.04	0.06	-0.02	0.01	-0.00	0.02
	Relative difference	-15.70	-14.32	-15.77	17.74	-9.62	-3.53	4.98	2.50	0.01	17.92	5.84	4.79	-40.74	-8.65	11.78	-2.35	6.12	-7.41	33.96
Cluster 2: 252 objects	Cluster mean	8.05	39.10	7.48	0.71	56.12	1.38	0.39	0.49	1.00	1.48	4.69	0.44	0.08	0.17	0.76	0.83	0.10	0.04	0.02
	Difference	-3.88	-19.41	-2.92	-6.67	-28.27	-0.05	-0.58	0.06	-0.00	-0.22	1.33	-0.00	0.03	-0.28	0.26	-0.01	0.01	0.02	-0.02
	Relative difference	-32.50	-33.18	-28.10	-90.40	-33.49	-3.29	-59.94	14.43	-0.20	-13.05	39.65	-0.12	50.79	-62.80	50.98	-1.03	15.93	80.38	-49.34

		trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	passenger_count	extra	mta_tax	improvement_surcharge	congestion_surcharge	Airport_fee	cbd_congestion_fee	Vendor			payment_type			
														Creative Mobile Technologies, LLC	Curb Mobility, LLC	Helix	Credit card	Cash	No charge	Dispute
Grand mean		11.93	58.52	10.41	7.38	84.39	1.43	0.97	0.43	0.99	1.7	3.36	0.44	0.05	0.45	0.50	0.84	0.09	0.02	0.05
Cluster 3: 195 objects	Cluster mean	18.79	90.55	17.70	12.11	125.98	1.63	1.49	0.34	1.00	1.26	1.19	0.40	0.08	0.89	0.04	0.92	0.04	0.01	0.04
	Difference	6.86	32.03	7.29	4.73	41.59	0.19	0.52	-0.09	0.00	-0.44	-2.17	-0.04	0.03	0.44	-0.47	0.08	-0.05	-0.02	-0.01
	Relative difference	57.46	54.74	70.01	64.14	49.29	13.52	54.12	-20.69	0.20	-26.09	-64.45	-9.04	53.85	98.03	-92.85	9.02	-53.90	-76.69	-23.62

Table A2. Cluster interpretation for K-Means with 7 clusters. The most outstanding results, referenced in the main text, are highlighted in green

		trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	passenger_count	extra	mta_tax	improvement_surcharge	congestion_surcharge	Airport_fee	cbd_congestion_fee	Vendor			payment_type			
														Creative Mobile Technologies, LLC	Curb Mobility, LLC	Helix	Credit card	Cash	No charge	Dispute
Grand mean		11.93	58.52	10.41	7.38	84.39	1.43	0.97	0.43	1.00	1.70	3.36	0.44	0.05	0.45	0.50	0.84	0.09	0.02	0.05
Cluster 0: 106 obj.	Cluster mean	4.41	23.13	1.87	3.01	35.25	1.46	0.70	0.50	0.99	1.44	3.30	0.34	0.15	0.31	0.54	0.52	0.31	0.11	0.06
	Difference	-7.52	-35.38	-8.54	-4.37	-49.14	0.03	-0.27	0.07	-0.01	-0.26	-0.06	-0.10	0.10	-0.14	0.04	-0.32	0.22	0.09	0.01
	Relative difference	-63.03	-60.47	-82.08	-59.16	-58.23	2.11	-27.96	15.18	-0.74	-15.37	-1.70	-23.51	202	-30.51	7.12	-38.38	250	415	20.43

		trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	passenger_count	extra	mta_tax	improvement_surcharge	congestion_surcharge	Airport_fee	cbd_congestion_fee	Vendor			payment_type			
														Creative Mobile Technologies, LLC	Curb Mobility, LLC	Helix	Credit card	Cash	No charge	Dispute
Grand mean		11.93	58.52	10.41	7.38	84.39	1.43	0.97	0.43	1.00	1.70	3.36	0.44	0.05	0.45	0.50	0.84	0.09	0.02	0.05
Cluster 1: 27 objects	Cluster mean	35.61	173.36	16.62	10.42	205.78	1.41	2.10	0.28	1.00	0.37	1.68	0.11	0.11	0.85	0.04	0.78	0.19	0.00	0.04
	Difference	23.68	114.84	6.21	3.04	121.39	-0.02	1.14	-0.15	0.00	-1.33	-1.69	-0.33	0.06	0.40	-0.46	-0.06	0.10	-0.02	-0.01
	Relative difference	198	196	59.65	41.22	143.85	-1.72	118	-35.40	0.20	-78.21	-50.14	-74.97	122	90.15	-92.62	-7.63	108	-100	-21.20
Cluster 2: 162 objects	Cluster mean	13.78	67.57	1.91	12.07	88.17	1.65	2.07	0.35	0.99	1.56	1.29	0.47	0.05	0.90	0.05	0.44	0.27	0.06	0.23
	Difference	1.85	9.06	-8.50	4.69	3.78	0.22	1.10	-0.08	-0.00	-0.14	-2.07	0.03	-0.00	0.45	-0.45	-0.40	0.18	0.04	0.19
	Relative difference	15.49	15.48	-81.65	63.50	4.48	15.09	114	-18.17	-0.42	-8.32	-61.48	6.36	-1.23	101.17	-90.16	-47.95	198	181	399
Cluster 3: 353 objects	Cluster mean	9.54	44.68	11.64	7.50	73.52	1.26	0.58	0.48	1.00	2.19	4.63	0.48	0.02	0.19	0.79	1.00	0.00	0.00	0.00
	Difference	-2.39	-13.84	1.23	0.12	-10.86	-0.17	-0.39	0.05	0.00	0.49	1.27	0.03	-0.03	-0.26	0.29	0.16	-0.09	-0.02	-0.05
	Relative difference	-20.02	-23.65	11.79	1.60	-12.88	-11.77	-39.87	11.67	0.20	28.73	37.79	7.19	-60.34	-57.63	57.44	18.76	-100	-100	-100

		trip_distance	fare_amount	tip_amount	tolls_amount	total_amount	passenger_count	extra	mta_tax	improvement_surcharge	congestion_surcharge	Airport_fee	cbd_congestion_fee	Vendor			payment_type			
														Creative Mobile Technologies, LLC	Curb Mobility, LLC	Helix	Credit card	Cash	No charge	Dispute
Grand mean		11.93	58.52	10.41	7.38	84.39	1.43	0.97	0.43	1.00	1.70	3.36	0.44	0.05	0.45	0.50	0.84	0.09	0.02	0.05
Cluster 4: 169 objects	Cluster mean	14.98	79.05	19.66	11.93	116.25	1.64	1.43	0.34	1.00	1.35	1.08	0.43	0.07	0.89	0.04	1.00	0.00	0.00	0.00
	Difference	3.05	20.54	9.25	4.54	31.87	0.21	0.47	-0.09	0.00	-0.35	-2.28	-0.01	0.02	0.45	-0.46	0.16	-0.09	-0.02	-0.05
	Relative difference	25.57	35.10	88.89	61.58	37.76	14.46	48.58	-20.19	0.20	-20.81	-67.92	-3.05	30.18	99.44	-91.75	18.76	-100	-100	-100
Cluster 5: 175 objects	Cluster mean	10.37	48.88	9.89	0.34	68.74	1.34	0.21	0.49	1.00	1.63	5.31	0.49	0.03	0.11	0.86	0.95	0.03	0.00	0.01
	Difference	-1.56	-9.63	-0.52	-7.04	-15.65	-0.09	-0.76	0.06	0.00	-0.07	1.94	0.05	-0.02	-0.33	0.36	0.11	-0.05	-0.02	-0.04
	Relative difference	-13.07	-16.46	-5.00	-95.40	-18.55	-6.23	-78.40	12.96	0.20	-4.20	57.84	11.00	-42.86	-74.49	70.75	13.34	-61.48	-100	-75.68
Cluster 6: 8 objects	Cluster mean	69.30	344	36.51	12.86	398	1.75	2.00	0.06	1.00	0.00	1.31	0.19	0.00	1.00	0.00	0.75	0.25	0.00	0.00
	Difference	57.37	285	26.10	5.48	313	0.32	1.03	-0.37	0.00	-1.70	-2.05	-0.26	-0.05	0.55	-0.50	-0.09	0.16	-0.02	-0.05
	Relative difference	481	488	251	74.29	371.41	22.21	107	-85.47	0.20	-100	-60.95	-57.77	-100	123	-100	-10.93	181	-100	-100